NITTE EDUCATION TRUST | NITTE MEENAKSHI INSTITUTE OF TECHNOLOGY

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Lesson Plan

Subject: *Artificial Intelligence* **Department:** *Artificial Intelligence & Data Science*

Semester: 4th

MODULE 1: **History of AI and state space search** Target Lecture Hours: 8	Suggested Reading materials/links: Textbooks: 1. A First Course in Artificial Intelligence, Deepak Khemani, McGraw Hill Education (India), 2013 2. Artificial Intelligence Structures and Strategies for Complex Problem Solving, George F Luger, Pearson. 6th Edition

Sl. No.	Date	Topics to be covered	No. of hours required	Co's	Blooms level	Actual date of Completion	Mode of teaching (board/ presentation, etc.)	Remarks
1	03/03/25	History, Can Machines think? Turing Test	1	CO1	L2	03/03/25	Both	Done
2	04/03/25	Winograd Schema Challenge, Language	1	CO1	L2	04/03/25	Both	Done
3	05/03/25	Philosophy, Mind, Reasoning, Computation, The Chess Saga	1	CO1	L2	05/03/25	Both	Done
4	06/03/25	Intelligent Agents State Space Search: Depth First Search	1	CO1	L2	06/03/25	Both	Done
5	07/03/25	Breadth First Search	1	CO1	L2	07/03/25	Both	Done
6	10/03/25	Depth First Iterative Deepening	1	CO1	L2	10/03/25	Both	Done
7	13/03/25	Heuristic Search: Best First Search, Hill Climbing	1	CO1	L2	13/03/25	Both	Done
8	13/03/25	Solution Space, Travelling Salesman Problem	1	CO1	L2	13/03/25	Both	Done

Planned Module 1 assessment date:

Actual Date of completion: 13/03/25

Total Strength of the class: 67

MODULE 2: **Search strategies and algorithms** Target Lecture Hours: 8 Target Tutorial Hours: 1		Suggested Reading materials/links: Textbooks: 1. A First Course in Artificial Intelligence, Deepak Khemani, McGraw Hill Education (India), 2013 2. Artificial Intelligence Structures and Strategies for Complex Problem Solving, George F Luger, Pearson. 6th Edition						
Sl. No.	Date	Topics to be covered	No. of hours required	Co's	Blooms level	Actual date of Completion	Mode of teaching (board/ presentation, etc.)	Remarks
1	14/03/25	Escaping Local Optima, Random Walk Algorithm	1	CO2	L2	14/03/25	Both	Done
2	17/03/25	introduction to Simulated Annealing, Population based methods	1	CO2	L2	17/03/25	Both	Done
3	19/03/25	Concepts of Genetic Algorithms and application to Eight Queens problem	1	CO2	L2	19/03/25	Both	Done
4	20/03/25	SAT, TSP, Emergent Systems	1	CO2	L2	20/03/25	Both	Done
5	20/03/25	Overview of Concepts of Ant Colony Optimization and application to TSP	1	CO2	L2	20/03/25	Both	Done
6	21/03/25	Overview of Concepts of Ant Colony Optimization and application to TSP	1	CO2	L2	21/03/25	Both	Done
7	24/03/25	Finding Optimal Paths: Branch & Bound	1	CO2	L2	24/03/25	Both	Done

8	26/03/25	A*, Admissibility of A*, Informed Heuristic Functions	1	CO2	L2	26/03/25	Both	Done

Planned Module 2 assessment date:

Type of assessment: assignment COs covered 1 & 2 and Bloom's level :2.

Actual Date of completion: 26/03/25

Total Strength of the class: 67

MODULE 3: **More on a* algorithm and game playing** Target Lecture Hours: 7	Suggested Reading materials/links: Textbooks: 1. A First Course in Artificial Intelligence, Deepak Khemani, McGraw Hill Education (India), 2013 2. Artificial Intelligence Structures and Strategies for Complex Problem Solving, George F Luger, Pearson. 6th Edition							
Sl. No.	Date	Topics to be covered	No. of hours required	COs	Blooms level	Actual date of Completion	Mode of teaching (board/ presentation, etc.)	Remarks
1	27/03/25	Results related to Admissibility of A*	1	CO3	L3	27/03/2025	Both	Done
2	27/03/25	Space Saving Version of A*: Iterative Deepening A*,	1	CO3	L3	27/03/25	Both	Done
3	28/03/25	Game Playing: Board Games and Game Trees	1	CO3	L3	28/03/25	Both	Done
4	23/04/25	Algorithm MiniMax,	1	CO3	L3	23/04/25	Both	Done
5	24/04/25	Algorithm AlphaBeta	1	CO3	L3	24/04/25	Both	Done
6	24/04/25	Limitations of Search; overview of examples	1	CO3	L3	24/04/25	Both	Done
7	25/04/25	Chess, Go and Backgammon	1	CO3	L3	25/04/25	Both	Done

Actual Date of completion: 25/04/25

Total Strength of the class: 67

MODULE 4: **First order logic and prolog**	Suggested Reading materials/links:
Target Lecture Hours: 7	Textbooks: 1. A First Course in Artificial Intelligence, Deepak Khemani, McGraw Hill Education (India), 2013 2. Artificial Intelligence Structures and Strategies for Complex Problem Solving, George F Luger, Pearson. 6th Edition

Sl. No.	Date	Topics to be covered	No. of hours required	Cos	Blooms level	Actual date of Completion	Mode of teaching (board/ presentation, etc.)	Remarks
1	05/05/25	Rule Based Expert Systems	1	CO4	L2	05/05/25	Both	Done
2	07/05/25	Knowledge, Reasoning and Planning	1	CO4	L2	07/05/25	Both	Done
3	08/05/25	First order logic	1	CO4	L2	08/05/25	Both	Done
4	08/05/25	Knowledge base;	1	CO4	L2	08/05/25	Both	Done
5	09/05/25	Application: A Logic based Financial Advisor	1	CO4	L2	09/05/25	Both	Done
6 & 7	12/05/25 & 14/05/25	Concepts of Programming using Prolog	2	CO4	L2	12/05/25 & 14/05/25	Both	Done

Planned Module 4 assessment date:

Type of assessment: assessment Bloom's level: 2.

Actual Date of completion: 17/06/2024

Total Strength of the class: 70

MODULE 5: **Planning**	Suggested Reading materials/links:
Target Lecture Hours:8	Textbooks: 1. A First Course in Artificial Intelligence, Deepak Khemani, McGraw Hill Education (India), 2013

		2. Artificial Intelligence Structures and Strategies for Complex Problem Solving, George F Luger, Pearson. 6th Edition						
Sl. No.	Date	Topics to be covered	No. of hours required	Cos	Blooms level	Actual date of Completion	Mode of teaching (board/ presentation, etc.)	Remarks
1	15/05/25	Planning Domain Definition Language and examples	1	CO5	L2	15/05/25	Both	Done
2	16/05/25	Transport, Blocks World,	1	CO5	L2	16/05/25	Both	Done
3	19/05/25	The complexity of classical planning	1	CO5	L2	19/05/25	Both	Done
4 & 5	21/05/25 & 22/05/25	Forward(progression) state space search,	2	CO5	L2	21/05/25 & 22/05/25	Both	Done
6 & 7	22/05/25	Backward(regression) relevant- states search	2	CO5	L2	22/05/25	Both	Done
8	23/05/25	GRAPHPLAN algorithm	1	CO5	L4	23/05/25	Both	Done
9	23/05/25	Means Ends Analysis	1	CO5	L4	23/05/25	Both	Done

Planned Module 5 assessment date:

Actual Date of completion: 03/07/2024

Total Strength of the class: 70

Plan of additional activities:

Learning Assessment 1: Assignment - 10 marks

Learning Assessment 2: Quiz1 & Quiz2 – 10 marks